

A NEW APPARATUS FOR ELECTROPHORETIC ANALYSIS OF COLLOIDAL MIXTURES.

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In the course of earlier work¹ on the electrophoretic migration of proteins in various buffer solutions (using a photographic-microphotometric method to follow the migration), a detailed study was made of the different sources of error in electrophoretic measurements and the possibilities of their elimination. Special interest was devoted to the electrophoretic behaviour of mixtures of proteins. It was found that the individual components in such mixtures do not usually mutually influence migration to any great extent, so that a partial separation is brought about by the current. From this we can derive conclusions as to the electrochemical homogeneity of a given colloidal solution, similar

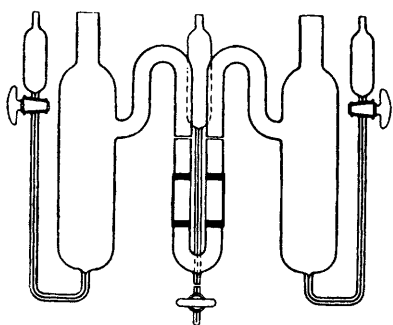


FIG. 1.—Electrophoresis apparatus for the moving boundary method (Tiselius, 1930).

to those Svedberg derives from his ultracentrifuge molecular weight analysis. The importance of using physical methods for such a purpose is obvious: the usual methods of fractionation of biocolloids (which are very unstable and very sensitive towards even quite mild chemical agents), are often of doubtful value. The procedure was exemplified both on artificial and on natural mixtures, and was also used for the purification of proteins. Fig. 1 shows the apparatus used.

The value of electrophoretic fractionation and purification of mixtures of colloidal systems was first realised by Botho Schwerin, who described an experimental arrangement and some applications of the method in a number of patents from 1914 onwards.² This work, of a preparative nature, was carried out in cells separated by permeable membranes. In addition to the author's work¹ on proteins, quantitative investigations (free electrophoresis with reversible electrodes) on mixtures have been carried out recently by Theorell,³ especially on respiratory ferments, and by Bennhold⁴ on the serum proteins and their reactions with

¹ A. Tiselius, "The Moving Boundary Method of Studying the Electrophoresis of Proteins," *Inaugural Dissertation Upsala*, 1930; *Nova Acta Reg. Soc. Scient. Upsaliensis*, IV., 7, No. 4. See also, Thé Svedberg, *Trans. Faraday Soc.*, 1930, 26, 737; and K. O. Pedersen, *Kolloid Z.*, 1933, 63, 268.

² For a review of this early work see the monograph *Elektrophorese, Elektro-osmose, Elektrodialyse in Flüssigkeiten*, Von P. H. Prausnitz and J. Reitstötter, Dresden u. Leipzig, 1931, especially pp. 120-122, 130, 138, 147-149.

³ H. Theorell, *Biochem. Z.*, 1930, 223, 93; 1934, 275, 1, 11, 19, 30; *Naturwiss.*, 1934, 22, 289-291; *Biochem. Z.*, 1935, 278, 291.

⁴ H. Bennhold, *Verhandl. deutsch. Ges. inn. Medizin. XLII. Kongress, Wiesbaden*, 1930, p. 353; *XLIII. Kongress, Wiesbaden*, 1931, p. 211; *Ergeb. inn. Medizin und Kinderheilkunde*, 1932, 42, 275; *Kolloid. Z.*, 1933, 62, 129.

other substances. Theorell also describes an apparatus with membranes and reversible electrodes for the electrophoretic purification of larger volumes of material. Electrophoretic methods provide a means of characterising the often difficultly distinguishable high molecular substances by isoelectric points and by p_H -mobility relationships, so as to get some idea of the degree of their electrochemical homogeneity, and to detect and more or less completely fractionate the homogeneous components in a mixture. It may also be possible, if suitable precautions are taken, to determine the relative proportions, and to study the degree, if any, of mutual combination of the components; in these two applications the tendency of the components to exert a mutual influence on the migration may interfere. For quantitative interpretation the experiments must be carried out at a salt concentration considerably higher than is usually needed for the depression of the boundary disturbances discussed previously.¹

The migration in the electrophoresis tube may suitably be followed by optical observation of the boundaries, or by analytical determination of the change of the quantities of the components above and below fixed levels in the tube, as used extensively for biocolloids by Hardy, Michaelis, Landsteiner and Pauli, and later by Reiner, Engel and Pauli, and Theorell.⁵ Both methods have their advantages. Optical observation is particularly useful when the differently migrating substances have a similar chemical composition (*e.g.*, mixtures of proteins) which cannot be separately determined analytically. On the other hand, the analytical method alone can be used for the study of enzymes and other materials, when a measurement of activity is the only means of determining where the substance is in the tube. The apparatus for the latter method differs mainly in the provisions for sampling.

The apparatus now to be described is adapted for simultaneous use of both methods. Whichever method is being used, the modifications in the experimental arrangement resulting from the following considerations are of importance, especially in the study of mixtures and in work in media of high conductivity, and in electrophoretic fractionation.

(1) Consider two components, A and B, with the mobilities u_A and u_B migrating in the same or in opposite directions in an electrophoresis tube of cross section area q cm.². The plus sign indicates a positive charge for u_A and u_B , and *vice versa*. If the current is i amp. and the conductivity κ ohm⁻¹, the potential gradient in the tube is $i/q\kappa$ volt/cm. and the boundaries will move with the velocities $iu_A/q\kappa$ and $iu_B/q\kappa$ cm. per sec. After a time t partial separation has taken place, so that, diffusion being neglected, the volume in c.c. on each side containing pure A or B is

$$\frac{i(u_A - u_B)tq}{q\kappa} = (u_A - u_B)\frac{it}{\kappa}.$$

Independently therefore of the dimensions of the apparatus, a given degree of separation corresponds to a certain amount of electricity sent through. It is assumed that reversible electrode tubes of some sort are used, and that the ions surrounding the electrodes have a mobility of u . When the quantity of electricity $i \times t$ is sent through, these ions similarly sweep through a volume in c.c. of $u \times \frac{it}{\kappa}$. If the volume in c.c. of colloidal solution

⁵ For literature references see, for example, Pauli-Valkó, *Kolloidchemie der Eiweisskörper*, 2nd ed., Chapter 10. Dresden u. Leipzig, 1933. For Theorell's work, *loc. cit.*³.

to be separated is v_s , there must be a volume of at least $\frac{u}{u_A + u_B} \times v_s$, between the electrodes and the boundaries, otherwise the ions originally present at the electrodes would reach the boundaries. In practice the volume should be about twice as large, since diffusion also interferes. If we desire to separate, say, 10 c.c. of two components with a difference in mobility of 0.5×10^{-5} , using reversible KCl(AgCl)Ag electrodes, the mobility of K and Cl being 65×10^{-5} , the calculated electrode vessel volume should be $\frac{65}{0.5} \times 10 = 1300$ c.c., and that used should be about 2600 c.c. The necessity of not only using reversible electrodes but also of allowing sufficient volume between the electrodes and the boundaries has often been overlooked in the construction of electrophoresis apparatus both for the moving boundary and for the transference method.

(2) When both substances migrate in the same direction the possibility of sufficient separation is limited by the fact that long before the desired separation has been reached, both substances have migrated out of the electrophoresis tube. This is a somewhat serious limitation, since the absolute differences in mobilities are very often much larger at p_H regions where both components have mobilities of the same sign (e.g. the serum proteins). Moreover, lack of solubility often prevents a choice of a p_H between the isoelectric points. For this reason we arranged for a slow and uniform movement of the solution in the electrophoresis tube at an exactly known rate and in a direction opposite to the migration, by slowly lifting a cylindrical glass tube by clockwork out of the liquid in one electrode tube during the electrophoresis. If the rate at which the tube is lifted is l cm. per hour, its cross-section area p cm.², the free surface of the liquid in each electrode tube Q cm.², and the cross-section area of the electrophoresis tube q cm.² then a movement of a given level in this tube will take place, at a rate of $\frac{lpQ}{q(2Q - p)}$ cm. per hour. By suitably choosing l and p

any desired rate can be obtained; even in the narrow tubes used in our apparatus a rate of several centimetres per hour did not markedly blur the boundaries in the electrophoresis tube.

For fractionation purposes, this "compensation movement" is adjusted so that the observed boundary separating two fractions obtains a suitable "apparent mobility" and consequently, at the end of the run, the column of the solution can be cut off exactly at the right place (see below).

(3) The advantage of using the highest possible potential gradient for getting good separation in the electrophoresis is obvious, since some substances show diffusion, which is appreciable when compared with the rate of electrophoretic separation brought about by the potential gradients commonly applied. However, a limit is imposed in this direction by the heat evolution, which causes thermal convection currents, gradually stirring up the contents.

This source of error has been examined in some detail¹ by sending alternating current through the apparatus instead of direct current, so that the heat effects could be studied separately. By observation of the boundaries the maximum allowable potential gradient V could be determined for given conditions. From this, and from the conductivity κ , the maximum allowable load w in watts per cubic centimetre in the electrophoresis tube could be calculated from the equation $w = V^2\kappa$. From this determination, for any given conductivity, the maximum allowable V could be calculated. For the apparatus earlier used¹ with a quartz U-tube of about 0.8 cm.² cross-sectional area and with protein solutions of about 0.5 per cent. strength, a maximum load of about 0.01-0.02 watts per cubic centimetre was found.

The convection currents set up with A.C. in tubes of varying shape and size, wall-thickness and so on, were studied in order to see if this maximum

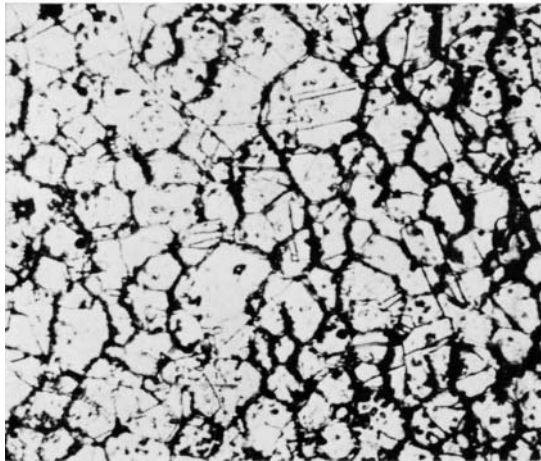


FIG. 1.—Macrocrystalline copper substrate. $\times 200$.

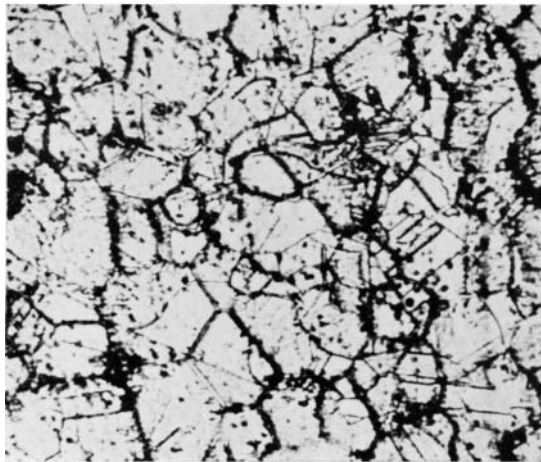


FIG. 2.—Nickel deposit, 5×10^3 A thick, deposited from strong bath at 0.6 A/dcm^2 . $\times 200$.

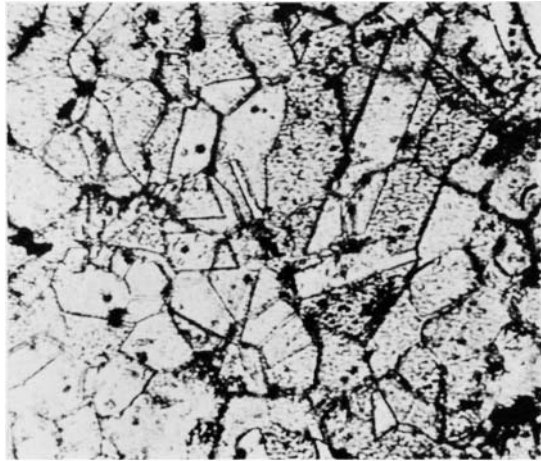


FIG. 3.—Nickel deposit, 2×10^4 A thick, deposited from strong bath at 1.2 A/dcm^2 . $\times 200$.

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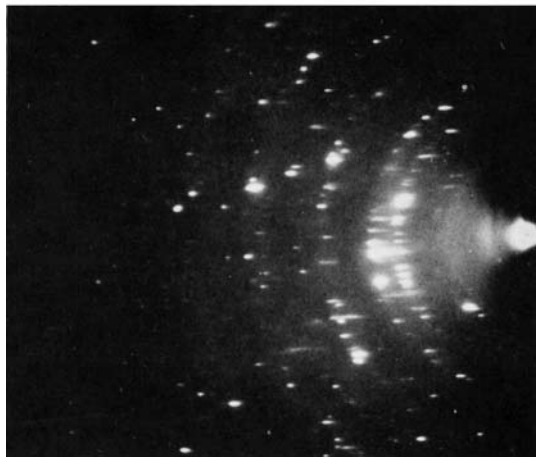


FIG. 4.—Pattern from macrocrystalline copper (see Fig. 1).

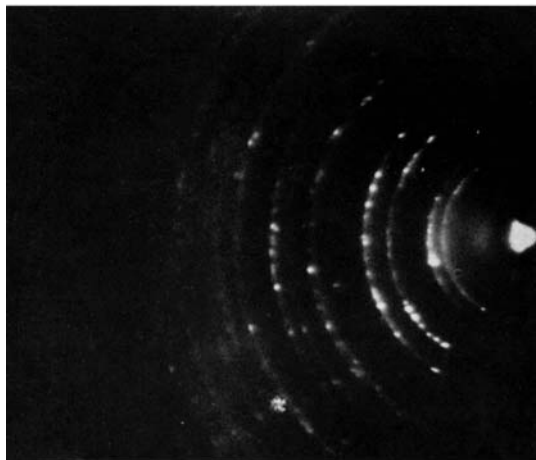


FIG. 5.—Pattern from nickel deposit of Fig. 2.

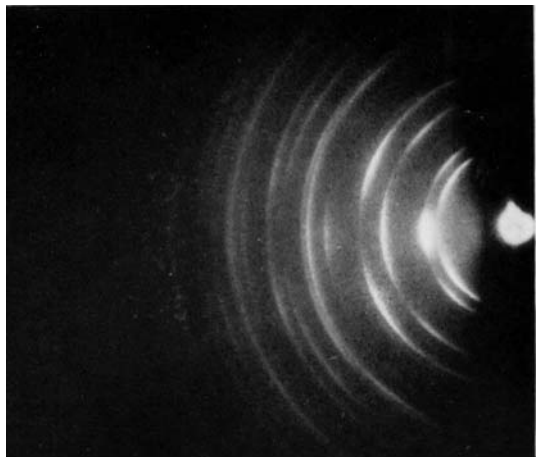


FIG. 6.—Pattern from nickel deposit of Fig. 3.
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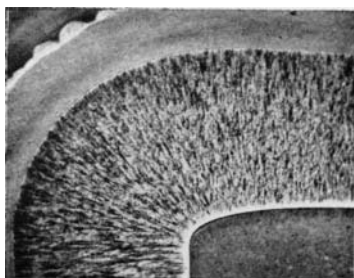


FIG. 1.—Electrolytic nickel from a nickel sulphate solution containing ammonium chloride. Structure along edge of cathode. $\times 75$.



FIG. 2.—Same deposit as Fig. 1, at $\times 250$.



FIG. 3.—Electrolytic nickel from a nickel sulphate solution containing sodium chloride. Structure along edge of cathode. $\times 75$.

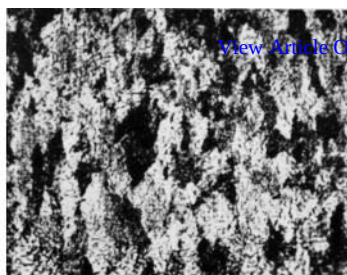


FIG. 4.—Electrolytic nickel from a nickel sulphate solution containing nickel chloride. $\times 250$.



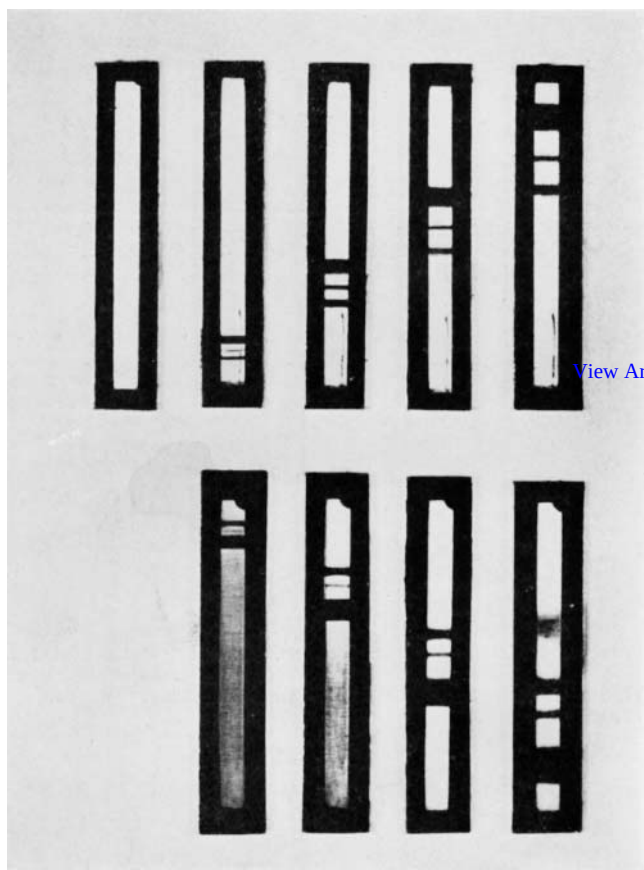
FIG. 5.—Cold drawn nickel rod. $\times 75$.



FIG. 6.—Structure of nickel rod after recrystallisation. $\times 75$.

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FIG. 4.—Electrophoretic analysis of blood serum. (The length of the cell is 40 mm.)

[See page 530.

load could be increased. Thermal convection, which invariably occurs in electrophoresis experiments at certain parts of the tube, becomes serious when the density of the central part of the solution beneath the boundary falls below that of the supernatant solution near the periphery, so that the difference in density due to the dissolved material, which keeps the boundary stable, is reduced to zero or less; at this point convection sets in, as shown in Fig. 2. However, as there is always some diffusion, layers of lower concentration above the boundary are upset by convection currents at much lower temperature differences than those at which the bulk of the solution is stirred up. This can be observed easily, when alternating current of increasing strength is sent through a U-tube with a boundary of some coloured substance.

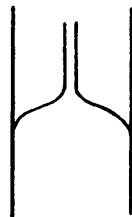


FIG. 2.—Thermal convection.

Experiment and theory both show that convection currents arise much more readily in wide than in narrow tubes. If the load is w watts/c.c., the thermal conductivity of the solution K and the temperature t , the temperature gradient dt/dr at a distance r from the centre of a cylindrical tube is $dt/dr = wr/2K$, and therefore proportional to the radius. This gives for the difference in temperature between the centre and the periphery of the cylindrical column

$$t_c - t_o = \frac{wr^2}{4K}.$$

It should therefore be very advantageous to use as narrow tubes as possible, were it not that such very small volumes of substance could then be handled. However it is possible to use *flattened* tubes: so that larger volumes can be accommodated with good heat conduction. In the apparatus to be described the electrophoresis tube has a rectangular cross-section of 3×25 mm., and in another apparatus under construction (for fractionating larger volumes) 5×100 mm. is used. With the A.C. procedure described above it was found that considerably higher load per cubic centimetre can be used than in a cylindrical tube of the same cross-sectional area; this is most marked for tubes of large cross-section.

Since thermal convection depends upon $d\rho/dt$, and water and the solutions usually studied have a density maximum at 4° C. or somewhat less, where $d\rho/dt$ is zero, it would seem that convection currents could be markedly suppressed by working at this temperature; this proved to be the case. Keeping the bath at about zero and using a load so high that the temperature in the tube is about 4° C., it was found, that, with the flattened tube, loads of 0.5-1.0 watts per cubic centimetre can very well be used; the potential gradients can therefore be about 10 times higher than before. Although the mobilities are smaller at the lower temperature, so that somewhat longer times have to be used, the diffusion being correspondingly reduced, the quality of the separation is not affected.

The advantage of using the optimum temperature was less marked for wider tubes, probably because when a large load is used, the temperature distribution of the solution is so wide that only part of the solution is at the optimum temperature.

Experimental Arrangement.

(1) The thermostat is a well insulated container taking about 100 l of water, and provided with double windows, between which are placed dishes with calcium chloride to prevent fog formation at the low temperature used. An Electrolux refrigeration unit of a maximum capacity of 100 kcal. per hour, but working at less than half-capacity, maintains the low temperature; the heat evolved in electrophoresis experiments usually does not exceed 20-30 kcal. per hour. Through a copper spiral wound around the evaporator of the refrigerator and connected to a similar spiral in the bath

by rubber tubing, alcohol was circulated by a small pump, driven by the same motor as the stirrer. This maintained the bath automatically at $+0.1-0.2^{\circ}\text{C.}$, even if the cooling effect varied within very wide limits. A compact layer of ice which is formed on the spiral in the bath is not removed by the stirring; if the cooling effect is large the thickness of this layer increases, but this makes the conduction of the heat to the spiral less efficient, so that the evaporator temperature drops somewhat and the real cooling effect decreases and *vice versa*. The arrangement, providing a constant temperature just above zero, would seem useful for many laboratory purposes, for example, dialysis.

(2) The electrophoresis apparatus (Fig. 3), consisted of a central U-tube

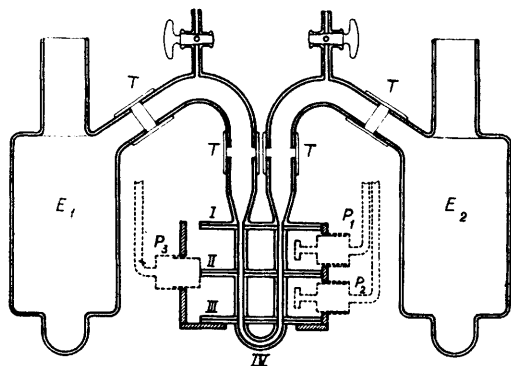


FIG. 3.—Apparatus for electrophoretic analysis.

E_1 , E_2 represent the electrode tubes (the larger type), T rubber tubings; I, II, III, IV, the four U-tube sections, of which II and III can be moved with the pneumatic arrangement P_1 , P_2 , P_3 .

connected by thick rubber tubing with the large volume electrode vessels of which there were two pairs, one of about 2 litres capacity each for very difficult separations, the other of about 0.5 litre each. Reversible AgCl electrodes with saturated KCl solution was used as in the older apparatus, but of course the electrodes were given much larger surface. The volume of KCl in each tube was 25-100 c.c.; it was allowed to form a layer at the bottom, running down through funnel tubes shortly before starting an experiment.

The U-tube, with an internal rectangular cross-section of $3 \times 25\text{ mm.}$, was made of plain parallel glass plates, cemented together with acid-proof silicate cement; the curved walls of the U-shaped bottom part were cut from an ordinary cylindrical glass tube of the right diameter.⁶

In order to follow the migration of any number of components analytically, the contents of the U-tube must be divided into at least two portions in one limb, preferably in both limbs. In most previous constructions (Cohn, Michaelis, Landsteiner-Pauli, Reiner, Engel-Pauli), this has been done with stopcocks. Bennhold takes out samples with a capillary pipette. Theorell divides the U-tube into sections between which ebonite plates with circular holes are moved, there being as many as seven sections in each limb for isolating fractions migrating in the same direction. With the compensation device described above, no more than two sections are needed, even for this purpose. When working with high voltage, very good insulation is necessary to prevent leakage of current into the thermostat from the U-tube; our U-tube, therefore, consists of two equal sections, with one U-shaped bottom part, and one top part (for connection to the electrode tubes), all cemented to large ($60 \times 120\text{ mm.}$) precision-ground glass plates, greased with a little vaseline thinned with paraffin oil. The arrangement resembles to a certain extent the diffusion apparatus of Cohen and Bruins.⁷ The four sections are held together in a metal stand with a slight pressure by means of rubber bands, so that they may be readily moved over each other. A movement of a little over 3 mm. of one section cuts it off com-

⁶ The glass apparatus was manufactured by the firm F. Hellige & Co., G.m.b.H., Freiburg in Breisgau, Germany.

⁷ Cohen and Bruins, *Z. physikal. Chemie*, 1924, **113**, 157.

pletely from the other. The sections are moved by the pneumatic arrangement (shown in Fig. 3 dotted lines), connected by rubber tubing to a small metal air pressure pump of the type commonly used for filling burettes, etc. This arrangement avoids vibration since necessary operations can be made without touching the apparatus. To secure exact alignment, the sections are pressed against a plane metal plate (to the right in Fig. 3). When making a run, the two lower sections are filled with the colloid, the U-tube "closed" by pressing section III. to the left, and the rest of the tube is then washed out with and finally filled with the buffer to be used, and connected to the electrode tubes. To save time, all solutions are kept in the cold room before filling the apparatus. The apparatus is left in the bath to attain constant temperature, after which the electrode vessels are levelled by connecting them for a few minutes through a tube between the two upper stopcocks. The clockwork to the compensation is then started, and the boundaries are made by moving section III. back again, after which the current may be applied. If optical study of the boundaries is to be made right from the start, the compensation may be allowed to work for some time before the current is started, so that the boundaries are brought out into the region where they can be observed. However, this is not usually necessary, since the boundaries soon become visible in any event. If two components are observed, their rate of migration is estimated, and from this a rate of compensation chosen, so that a value corresponding to the arithmetic mean (the mobilities taken with their signs) will not move at all. Then one boundary will move downwards (relatively to the tube) at exactly the same speed as the other moves upwards, and at the end of the experiment, the two sections to the left will contain only one component, the two to the right only the other. In this way the separation capacity of the apparatus can be utilised to its full extent even though both components migrate very rapidly in the same direction. At the end of the experiment sections II. and III. are moved slightly to the left, the apparatus is taken out, the electrode tubes removed, and the contents of the sections can be pipetted out with capillary pipettes, without taking the sections apart. Each section holds 4 c.c. so that 8 c.c. of solution of each component can be obtained. In the larger apparatus, with 5×100 mm. cross-section, this volume will be increased to 100 c.c., but it will probably not take quite so high a voltage, on account of the wider tubes. If it is only desired to measure the mobilities it is not necessary to run the boundaries so far. The mobility can be determined by observation of the boundaries or by analysis of the contents of the sections in the usual way.

(3) Observation of the moving boundaries of colourless substances cannot be followed so readily by ultra-violet photography as in the earlier apparatus, as this would necessitate the expense of using quartz both for the U-tube and for the large windows. The Topler "*schlieren*" projection method has, however, been adopted with quite good results. In the usual arrangement ^a a Dallmeyer objective of f about 60 cm. with large aperture (10 cm.) is placed as near to the apparatus as possible. A lamp and a diaphragm are placed at about 100 cm. distance, on the same side of the thermostat. The image of the diaphragm is thrown on the other side of the apparatus, at about 100 cm. from the window. Here a projection lens is placed, partially covered by a vertically movable screen with a sharp edge. Instead of the projection lens a camera objective can be used; in this way the photographs (Fig. 4) were obtained.

This method has proved much superior to the earlier light absorption method, especially when studying mixtures. With one and the same cell thickness, moreover, any concentration can be used, instead of the very narrow limits of concentration which the earlier method permits. By suitable adjustment of the edge in front of the lens any gradient can be observed, and even very diffuse boundaries may be located accurately.

There are several ways in which the amounts of the different components present may be determined optically, if the boundaries are well separated,

so that not only qualitative but quantitative electrophoretic analysis may be effected. In the light absorption method the photographs were microphotometred, and the concentrations computed from the photometric diagrams; the same procedure can be applied to the photographs obtained by the present method. The concentrations are obtained by graphic integration of the curves obtained.* The refractive index method originally suggested by the author and worked out by Lamm⁹ for use in the ultra-centrifuge and diffusion measurements may also be applied for the same purpose. For measurement of the migration of the boundaries, however, the above method is much more convenient.

(4) The potential source consists of a centre-tapped 2×1500 volt transformer, taking 100 m.a., with two half-wave mercury rectifying tubes of maximum peak inverse voltage of 7500 volts and maximum peak plate current 600 m.a. The filter circuit consists of a 20 Henry choke and a $4 \mu\text{F}$ condenser. The voltage fluctuations were small as the primary voltage was kept constant with an induction regulator. As in previous work current rather than voltage was measured, and a coulometer should be used for very accurate work especially if the voltage shows fluctuations.

Some Examples of the Application of the Apparatus.

(1) *Electrophoretic analysis of a mixture of R. phycocyan and R. phycoerythrin.*—These two proteins, which occur together in certain algæ, have been studied electrophoretically in detail by the author,¹ and by Pedersen:¹⁰ the isoelectric points are 4.85 and 4.25 and the slope of the mobility ϕ_H curve at the isoelectric points $du/d\phi_H = 10.2 \times 10^{-5}$ and $14.2 \times 10^{-5} \text{ cm.}^2 \text{ sec.}^{-1} \text{ volt}^{-1}$ at 20° C. Thus the separation is most favourable in alkaline solutions where the difference in mobility is about 10×10^{-5} , both migrating in the same direction. Separation at such a considerable difference in mobility is a very easy matter, especially since the proteins can be dissolved in buffer solutions of low conductivity, allowing a very high voltage to be used. Since the separation of these proteins can otherwise be performed only by repeated crystallisation (frequently with considerable loss of the phycocyan which usually amounts to only 10 per cent. of the phycoerythrin), this method has considerable preparative value and has been so used in our laboratory with very good results. In one experiment the voltage was 880 volts (23.2 volts per cm. in the tube), the current 15.5 milliamp., the substances were dissolved in phosphate buffer $m/100$ ϕ_H 6.96 of conductivity 0.77×10^{-3} at 0° C. , the heat effect was therefore $(23.2)^2 \times 0.77 \times 10^{-3} = 0.415$ watts per c.c. well below the maximum. The compensation was adjusted so that both boundaries migrated in opposite directions at the same rate. After two hours and twenty minutes the separation was completed: the cells on one side containing only phycocyan, those on the other only phycoerythrin. In separations with apparatus of the U-tube type one can of course never make use of the migration in the lowest U-shaped section which, being a dead volume, should be made relatively as small as possible: in our construction about 4.5 c.c.

(2) *Electrophoretic analysis of blood serum.*†—In this case the conductivity is rather high, so that the fact that the apparatus takes large currents is especially useful. The serum was dialysed against an phosphate buffer of ϕ_H 8.06, the same buffer being used as supernatant fluid. A voltage of 7.25 volts per cm. was used. Fig. 4 shows a number of photographs of the

* For a discussion of the quantitative application of the "schlieren" method in this manner see the work of Schardin.⁸

⁸ See the usual handbooks of Practical Physics. A recent survey: H. Schardin, *Das Toeplersche Schlierenverfahren*, VDI-Verlag, Berlin, 1934.

⁹ O. Lamm, *Z. physikal. Chemie*, A, 1928, 138, 313.

¹⁰ K. O. Pedersen, *Unpublished measurements*.

† Full details about the application of the apparatus for this purpose will be given in a forthcoming publication.

upwardly migrating boundaries on the positive side (upper row) and the downwardly migrating boundaries on the negative side (lower row), taken at time intervals of twenty minutes. The fastest boundary is serum albumin, the other two represent serum globulin fractions of different mobilities. At least one more component becomes visible if the screen is raised a little (see the last photograph in the lower row). In this experiment a compensation movement of 7.7 mm. per hour was used in order to bring out the slowest boundary of the region where it is screened off by the glass plates (lower part of cell in upper row). By suitable compensation the experiment can be arranged so that the different components can be isolated.

Summary.

Some improvements in the experimental arrangement for electrophoresis are discussed, with special attention to the study of mixtures. It is shown that a very considerable increase in the applied potential gradient can be obtained by changing the shape of the U-tube, and, above all, by keeping the electrophoresis tube at a temperature in the neighbourhood of the density maximum for water ($+4^{\circ}\text{C.}$), thus greatly reducing the risk of heat convection currents.

A modification of an earlier construction of electrophoresis apparatus is described, in which regard has been given to these points. The new apparatus has also provision for taking out samples of the separated fractions. The optical observation of the boundaries is made by an improved method depending upon refractive index (Töpler's *schlieren* method). For difficult separations large electrode vessels can be used, and a compensation arrangement, making it possible to obtain a large separation volume even if the components migrate in the same direction.

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